
UCSD ECE 285 Homework 2: VAE for Chest X-Ray Image Generation

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1 Task and Setup

I train a convolutional Variational Autoencoder (VAE) to generate grayscale chest radiographs from the Kaggle Chest X-Ray Pneumonia dataset (1). I report: (i) hyperparameters; (ii) training loss curves; (iii) FID/IS using the PyTorch-Ignite evaluation workflow (2); and (iv) examples of model outputs.

Data. I use the `train` split; images are converted to grayscale and resized to 128×128 .

Model. Let $x \in \mathbb{R}^{H \times W}$ and $z \in \mathbb{R}^d$. The encoder outputs $(\mu(x), \log \sigma^2(x))$, and I sample $z = \mu + \sigma \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$ (3). The decoder maps z to $\hat{x}$. I optimize the β -VAE objective (4):

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(z|x)}[\mathcal{L}_{\text{rec}}(x, \hat{x})] + \beta D_{\text{KL}}(q_\phi(z|x) \parallel \mathcal{N}(0, I)). \quad (1)$$

2 Hyperparameters and Evaluation Protocol

Common. $d = 64$, 128×128 , batch size 64, Adam learning rate 2×10^{-4} .

Run 1 (baseline). 15 epochs, BCE reconstruction, constant $\beta = 1.0$. Outputs: `loss1.png`, `samples_ep015.png`.

Run 2 (improved). 30 epochs, ℓ_1 reconstruction, KL warm-up: β increases linearly from 0 to $\beta_{\text{max}} = 1.0$ early, then stays at 1.0. Outputs: `loss2.png`, `samples_ep030.png`.

FID/IS. For each run, I generate $N = 5000$ samples by sampling $z \sim \mathcal{N}(0, I)$ and decoding. I compute FID and IS between 5000 generated samples and 5000 real images sampled from the training split using PyTorch-Ignite (2). Since IS uses ImageNet Inception features (5) and radiographs are out-of-domain, IS is expected to be near 1; I treat FID as the primary indicator (6).

3 Results

Loss curves. Figure 1 compares training dynamics. Run 1 shows a near-zero KL term (posterior collapse). Run 2 yields a sustained KL contribution after warm-up.

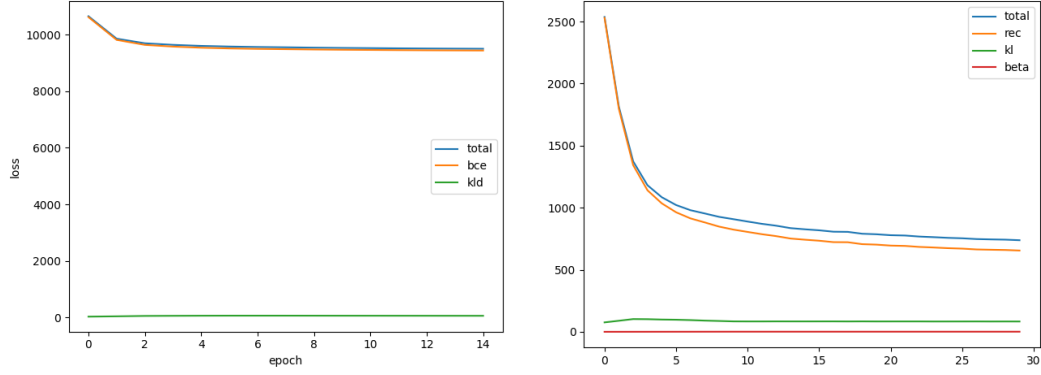
FID/IS. Table 1 reports evaluation on 5000 generated samples per run; Run 2 improves FID and slightly increases IS.

Generated examples. Figure 2 shows reconstructions (real on top, reconstructions on bottom). Run 2 better preserves global structure.

4 Discussion

Run 1 exhibits posterior collapse (near-zero KL) and overly smooth reconstructions. Run 2 (KL warm-up + ℓ_1 reconstruction) produces stronger latent utilization and better sample quality, reducing FID from 422.983 to 364.914 while keeping IS near 1 as expected in this domain. Overall, simple schedule/loss changes improve VAE training stability and generation quality on chest radiographs.

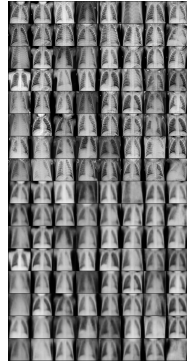
Preprint.



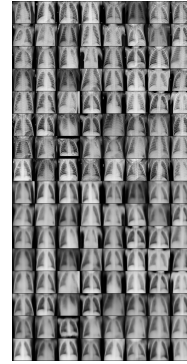
(a) Run 1: BCE + constant β .

(b) Run 2: ℓ_1 + KL warm-up.

Figure 1: Loss decomposition for Eq. (1) under two configurations.



(a) Run 1 at epoch 15.



(b) Run 2 at epoch 30.

Figure 2: Qualitative examples: real images (top) and VAE reconstructions (bottom).

References

- [1] P. Mooney. Chest X-Ray Images (Pneumonia). Kaggle dataset.
- [2] PyTorch-Ignite Blog. GAN evaluation with FID and IS. Referenced evaluation pipeline.
- [3] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. ICLR, 2014.
- [4] I. Higgins et al. β -VAE: Learning Basic Visual Concepts with a Constrained Variational Framework. ICLR, 2017.
- [5] C. Szegedy et al. Rethinking the Inception Architecture for Computer Vision. CVPR, 2016.
- [6] M. Heusel et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. NeurIPS, 2017.

Run	FID ↓	IS ↑	#Gen
Run 1 (baseline)	422.983	1.043 ± 0.002	5000
Run 2 (improved)	364.914	1.060 ± 0.001	5000

Table 1: FID and IS on 5000 generated samples per run (PyTorch-Ignite (2)). Lower FID indicates closer alignment to real images in Inception feature space (6). IS is mean $\pm$ std over splits.